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Working Group VIII - Satellite Galaxies

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Summary

This working group will study the formation and evolution of satellite galaxies as a function of various host halo and satellite properties, and across different numerical treatments of feedback and reionization.

The discussion forum for this working group can be found here:

https://groups.google.com/forum/#!forum/agora_satellites

Scientific Drivers

Our aim will be to address some of the pertinent open issues in this field, such as:

- I) What are the relative impacts of environmental processes, such as gravitational tides, ram pressure stripping and ionizing background radiation, versus internal processes, such as supernovae feedback, on the formation and evolution of dwarf galaxies?
- II) How the central DM density profiles of satellites depend on satellite luminosity and star formation history, and the possible formation of DM cores in low-luminosity satellites. This will address how varying resolution, and the different feedback implementations used in the participating codes, effect possible DM core formation.
- III) We will study the mass-loss rates due to tidal stripping for satellites with different orbital histories, infall masses, and DM density profiles. Particular attention will be paid to the role of the host disk ("disk shocking"), and the shape of the satellites' DM profiles on satellite survivability, and its implications for the missing satellite and "too big to fail" problems.
- IV) We will study the number of low-mass halos formed, and amount of baryons accreted onto such halos, before reionization ("pre-reionization fossils"), to determine: a) how many of these low-mass fossils host luminous satellite galaxies today b) whether pre-ionization fossils can make up some fraction of the MW's ultra-faint dSph population, and c) how this population is effected by different feedback and UV-background implementations.
- V) We will examine how all of these satellite properties, as well as the luminosity function of satellites, depend on host halo mass and environment.

Members (As of November 2012)

Javiera Guedes, Mike Boylan-Kolchin, Mike Kuhlen, Piero Madau, Lucio Mayer, Annalisa Pillepich, Joel Primack, Justin Read, Miguel Rocha, Christine Simpson, Adi Zolotov

Participating Codes

Below is a list of the participating codes, and the person who will serve as the contact person for that code in this working group. This list is very much *in progress*. I have contacted some folks from ART and Ramses and asked if they would like to join the group. **Please let me know if you can/would like to be a contact person.** For now, I've suggested a few people below.

Code	Contact Person in WG VIII
AREPO	?

ART – NMSU	
ART – Chicago	
ENZO	Christine Simpson
GADGET	
GADGET –SPHS	Justin Read (?)
GASOLINE	Adi Zolotov + Annalisa Pillepich (?) +
RAMSES	

Potential Analysis

Coming soon.

Progress & Results

not yet...